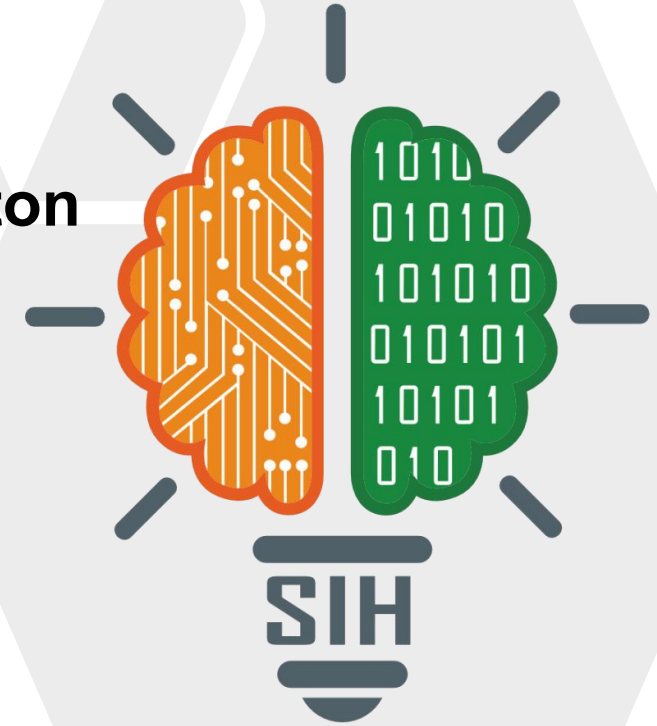
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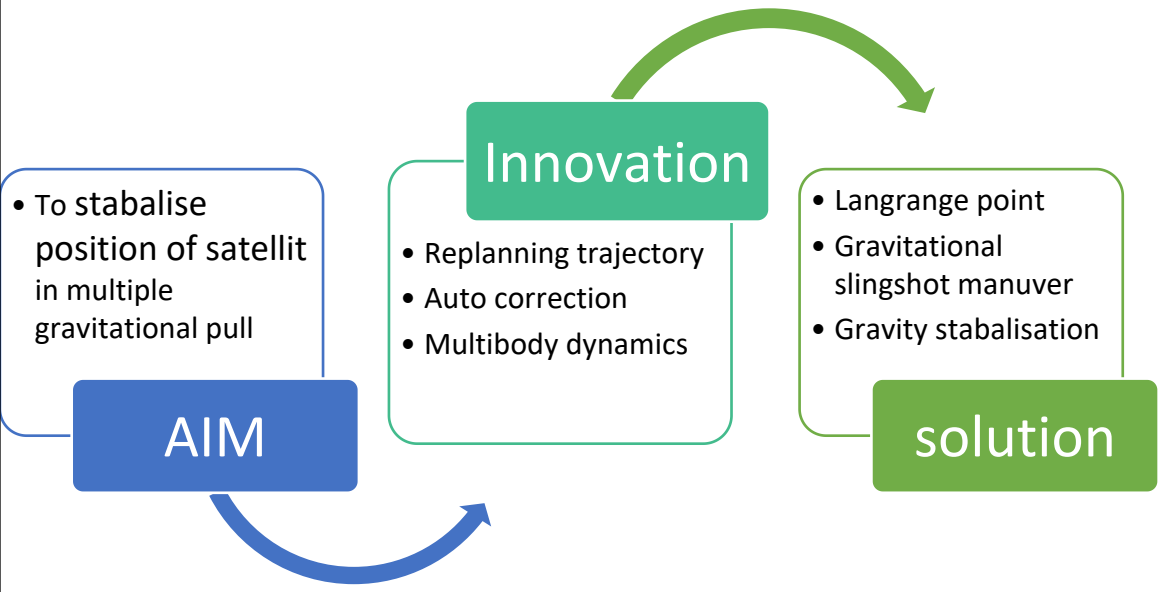
- **Problem Statement ID: – SIH 25142**
- **Problem Statement Title:- student innovaton**
- **Theme:- space technology**
- **PS Category:- Software**
- **Team Name :- nova_1**
- **Team id :- 52369**

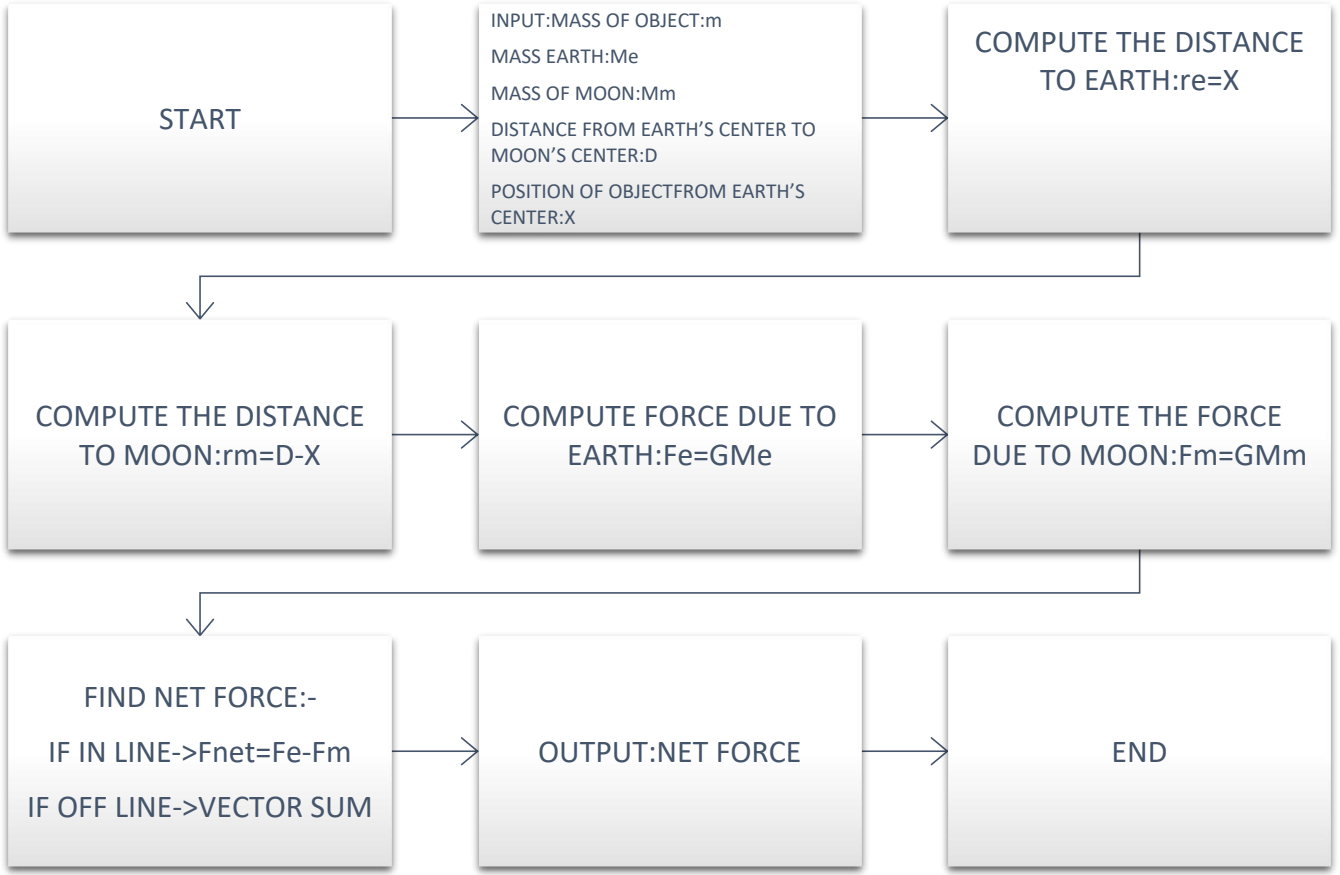


TEAM NOVA_1

Balancing objects in space with gravitational force

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TEAM NOVA_1

TECHNICAL APPROACHES

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Programming languages

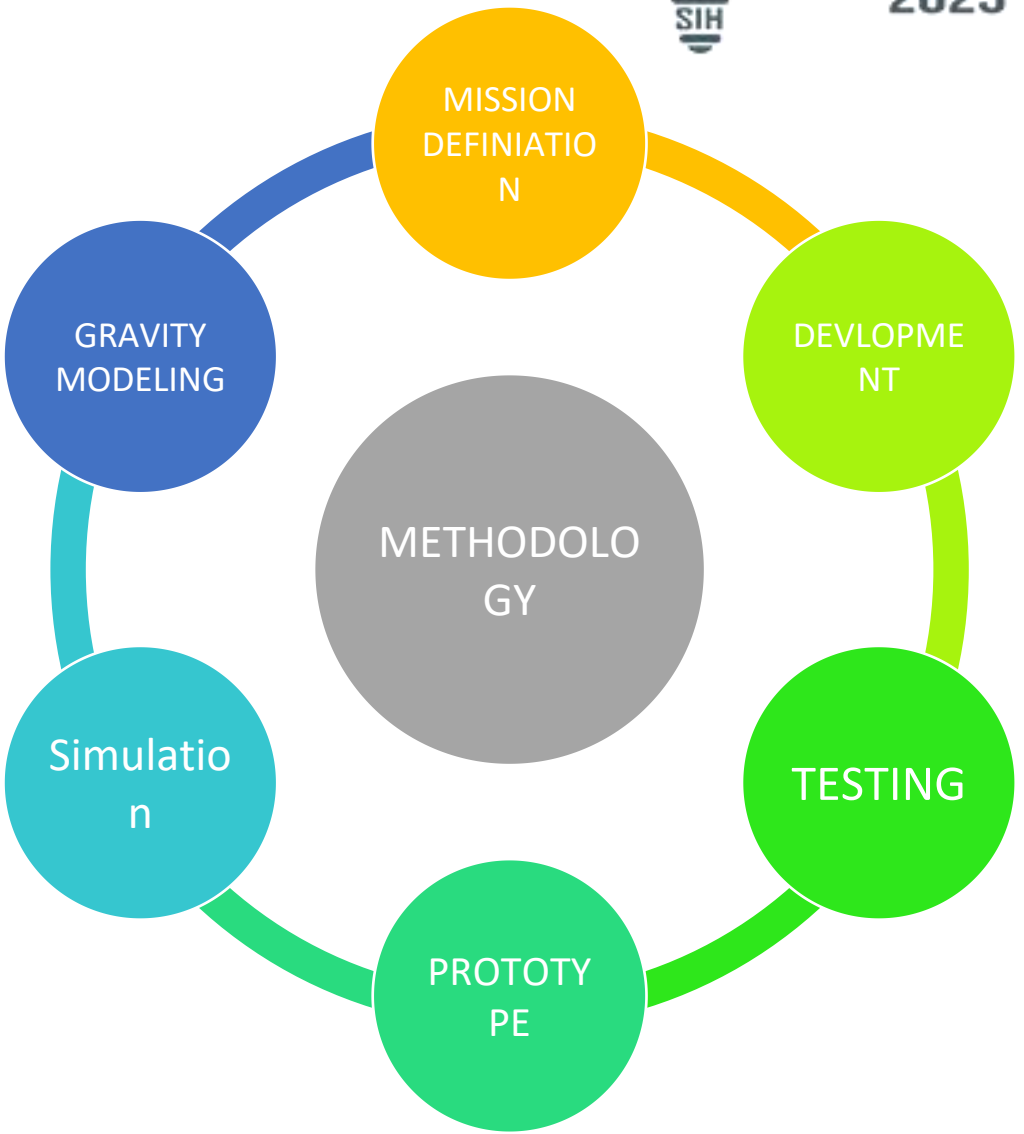
1. C / C++
2. PYTHON
3. MATLAB
4. TENSORFLOW
5. JAVA SCRIPT

FRAMEWORKS & TOOLS

01 JUPYTER NOTEBOOK

02 NumPY / SciPY

03 GMAT OR STK



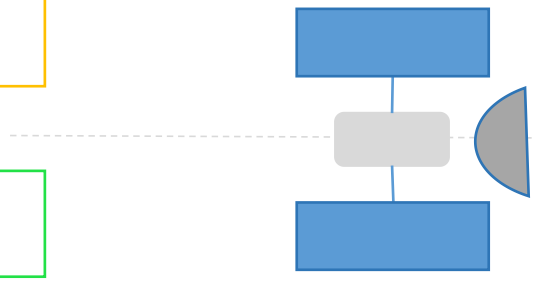
TEAM NOVA_1

FEASIBILITY AND VIABILITY

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Challanges & risk

- Thruster limit
- sensor noise
- Disturbance in computation



Solution

- Guidance navigation
- Digital twin code
- Multibody simulation

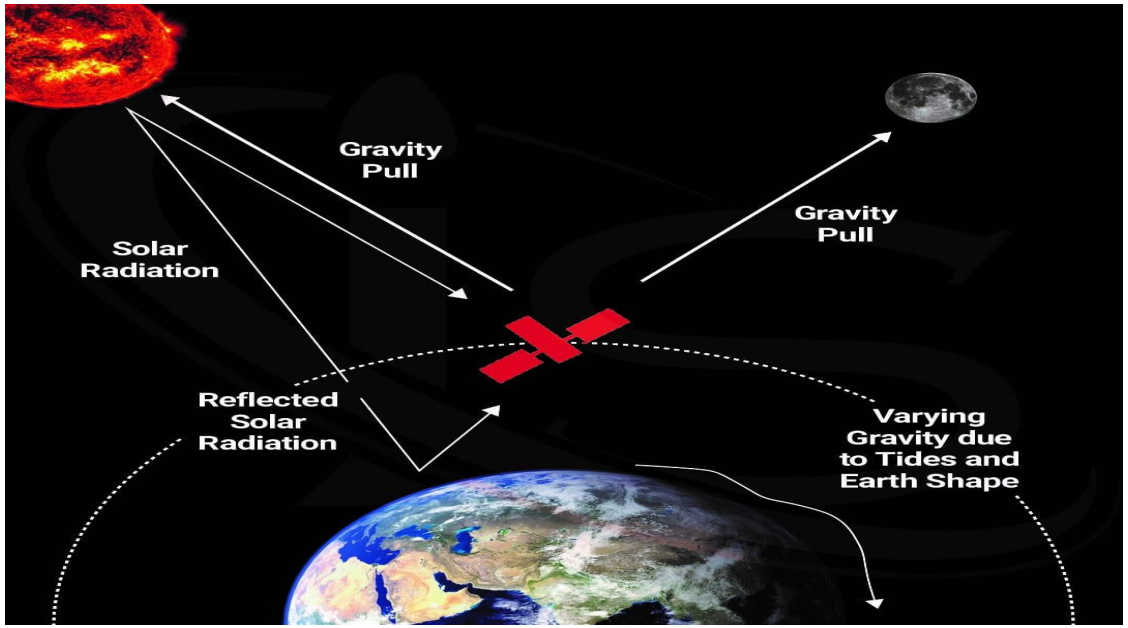
TEAM NOVA_1

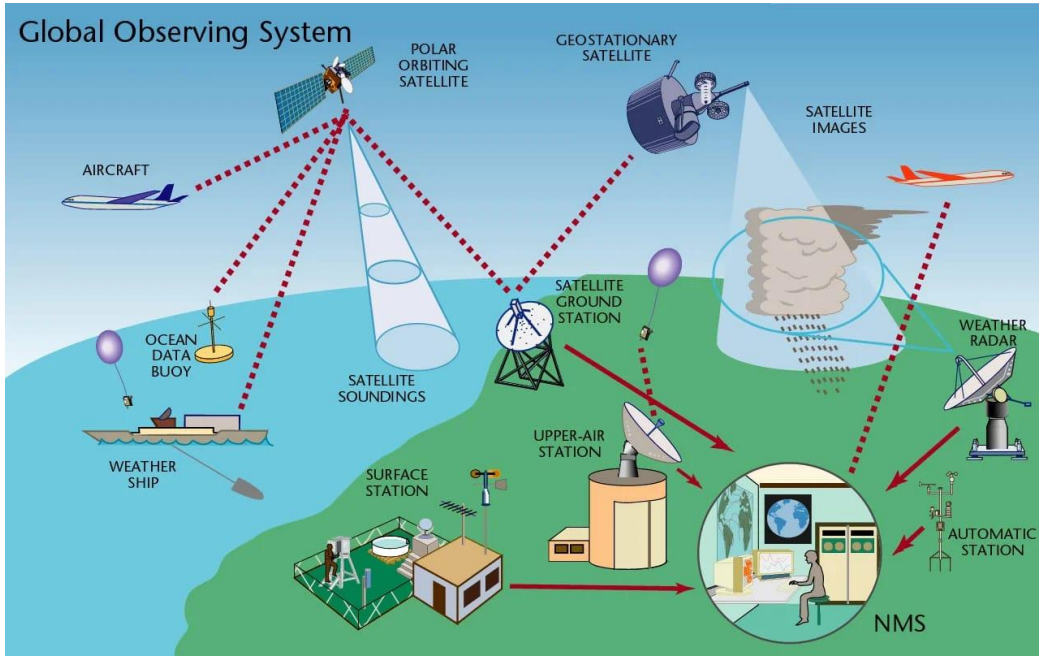
IMPACT AND BENEFITS

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❖ IMPACT

1. Satellite is pulled by Sun, Earth, and Moon at the same time.
2. Position becomes very sensitive to small forces.
3. Needs accurate balance and monitoring.
4. Once fixed, requires less fuel to stay stable.





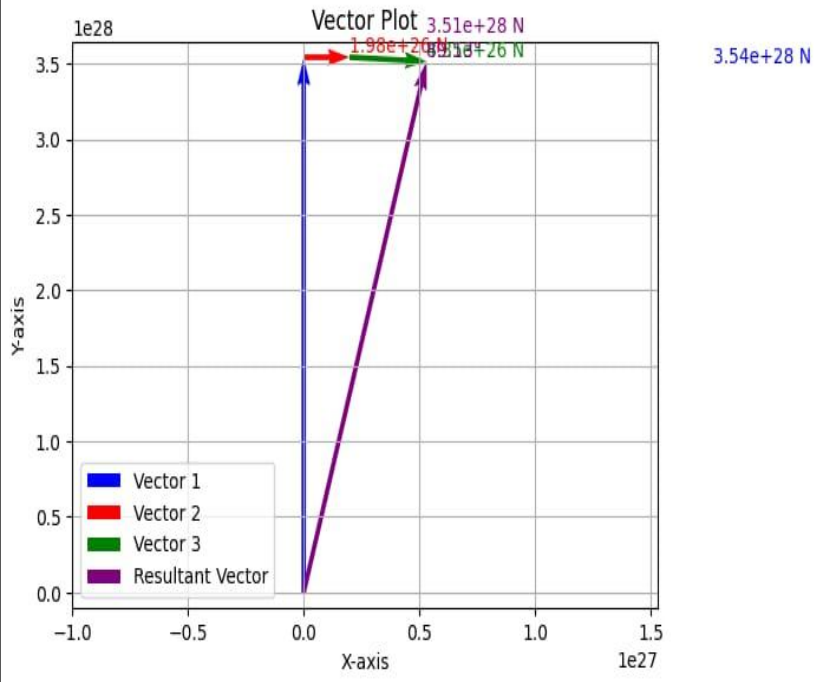
❖ Benefits

1. Increases satellite life by saving fuel.
2. Continuous observation of Sun, Earth, and Moon.
3. Supports scientific research and data collection.
4. Improves space communication and navigation.
5. Useful for future space missions and lunar projects.

TEAM NOVA_1

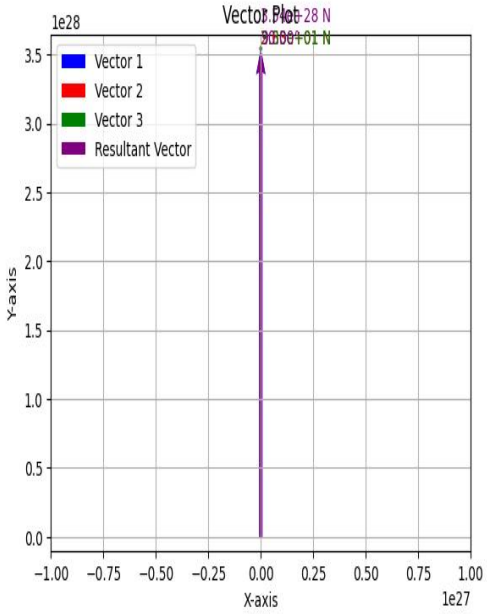
ORBITAL DYNAMICS PROTOTYPE

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3.54e+28 N





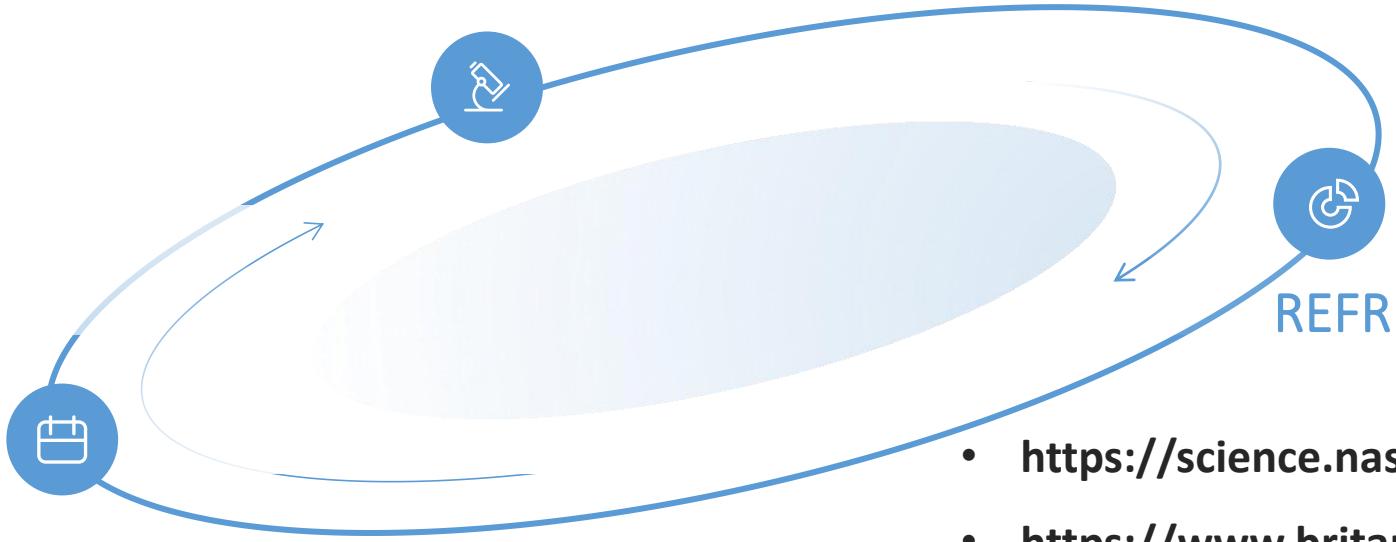
3.54e+28 N

- ❑ Graph 1 (Earth–Moon + Earth–Sun + Sun–Moon combined)
  - ➢ The resultant force is 3.54 × 10^{28} N, tilted just 1° from vertical.Conclusion: The Earth–Sun gravity dominates, while the Moon's forces only add a tiny sideways nudge.
- ❑ Graph 2 (with given vector magnitudes and angles)
  - ➢ The resultant again comes out as 3.54 × 10^{28} N, almost vertical. Conclusion: Even when extra vectors are introduced, the Sun–Earth pull overpowers, keeping the net force straight upward.
- ❑ Graph 3 (Moon at 90° case)
  - ➢ The resultant force is 3.54 × 10^{28} N, exactly 90° vertical. Conclusion: Here the Moon's forces cancel out sideways, leaving the Earth–Sun force to stand unchallenged.
- ❖ In all scenarios, the Earth–Sun gravitational pull rules the system. The Moon's effects are minimal—causing at most a 1° tilt—so to balance an object in space, thrust must primarily counter the Earth–Sun pull.

RESEARCH AND REFERENCES

RESEARCH

- Newton's law
- Langrange point
- Centrifugal force
- Tidal effect



APPLICATION

JWST

SOHO & DSCOVR

REFRENCE

- **https://science.nasa.gov/**
- **https://www.britannica.com/science/space-exploration**
- **https://science.nasa.gov/mission/new-horizons/**